

Programming Project Description – Roam

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27.05.2026

Many people use services such as Google Earth Street View daily as a tool to explore places around the world. Whether it's to look at a mountain trail to plan a hike, looking at a city to travel to, virtually touring a university, or even seeing how an area changes over time, street-level imagery gives users an immersive, first-person perspective on the world. However, most of these services rely on data collection from users through a web browser or mobile app, and do not open-source their imagery. Additionally, these services require having a modern web browser, which hardcore GNU/Linux users living in the terminal do not agree with.

There exists one service called Mapillary (<https://www.mapillary.com/>) that does open-source their street-level imagery under a Creative Commons Share-Alike license (<https://creativecommons.org/licenses/by-sa/4.0/>). Anyone can add their own imagery, even from a smartphone as the entire platform is crowd-sourced. Mapillary even provides an Application Programming Interface (API) to get access to imagery data and download the source images (<https://www.mapillary.com/developer/api-documentation>).

Using this service and interface, we aim to make street-level imagery accessible to all, even developers living inside a GNU/Linux terminal for most of their lives. To achieve this, we will query the user for a location, send HTTP GET requests to Mapillary's API to get the street-level imagery, convert spherical 360° images to perspective projection, extract pixel data from the images, represent the images as colored ASCII in the terminal, and handle user input for movement/navigation.

To the right is a block-diagram detailing the different parts to be used in this project. The entire project will be programmed in ANSI C99, and may use some other libraries also programmed in C99.

This project could be used as an art installation, as a proof-of-concept of what programming can do, and as a foundation for other interactive projects, such as a terminal-based cycling simulator, or a terminal-based 3D game based on real imagery.

To our knowledge, no other implementation of a street-level imagery viewer exists in the terminal. We kindly request the approval of this project to enable people to explore the world from their terminal!

